

Chapter-05

Distribution Function

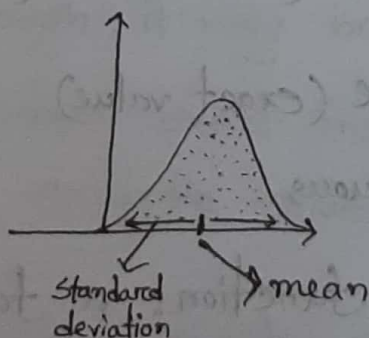
Cumulative Distribution Function: It is a function which gives the probability of a random variable being less than or equal to a given value. Denoted by $P(x_i)$ which implies that x takes the values less than or equals to x_i .

* Distribution of data

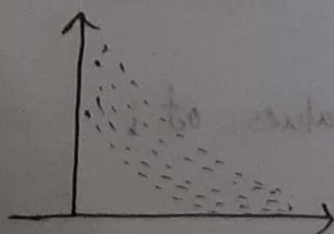
* Data structures; function for the structure

Data Properties:

i) Center limit theorem: center - a combined



ii) Skewed: creates asymmetrical, skewed curve



Examples of Distribution function -

i) Gamma distribution

ii) Gaussian

iii) Poisson

** Applications of Distribution

Algorithm $\rightarrow$ 2 types:

1. Direct input to output $\rightarrow$ Deterministic

2. বার বার randomly run করলে $\rightarrow$ Non-deterministic

System $\rightarrow$ 2 Types:

1. Deterministic

2. Non-deterministic (Stochastic - যা randomness maintain করে)

$\hookrightarrow$ i) Discrete (exact value)

ii) Continuous

Discrete probability function: a function that can take a discrete number of values (not necessarily finite).

Properties:

i) $P(x_i) \geq 0$ for all values of i

ii) $\sum_{i=1}^n P(x_i) = 1$

Chapter 01

Basic Simulation Modeling*** The Nature of Simulation**

Assumption $\rightarrow$ usually take the form of mathematical or logical relationships, constitute a model that is used to try to gain some understanding of how the system behaves.

$$y = mx + c$$

This can be a model. Because there is an assumption or mapping between the dependent and independent variable, and there's a relation between them.

Analytical solution: If the relationships that compose the model are simple enough it may be possible to use mathematical methods to obtain exact information on questions of interest; this is called analytic solution.

System too complex $\rightarrow$ simulation

**** Applications of Simulation**

**** Problems or impediments of simulation:**

1. Models used to study large-scale systems tend to be very complex and writing computer programs to execute them can be a difficult task.

⇒ This problem has been addressed by the development of excellent software products that automatically provide many of the features needed to program a simulation model.

2. Large ~~the~~ amount of computer time is required sometimes

⇒ This difficulty has become much ~~cheaper~~ less severe as computers become faster and cheaper.

3. There appears to be an unfortunate impression that simulation is just an exercise in computer programming that too a complicated one.

⇒ Consequently, many simulation studies have been composed of heuristic model building, programming and a single run of the program to obtain the answer.

Need design and analysis of simulation models, methodology carefully.

* Systems, Models and Simulation

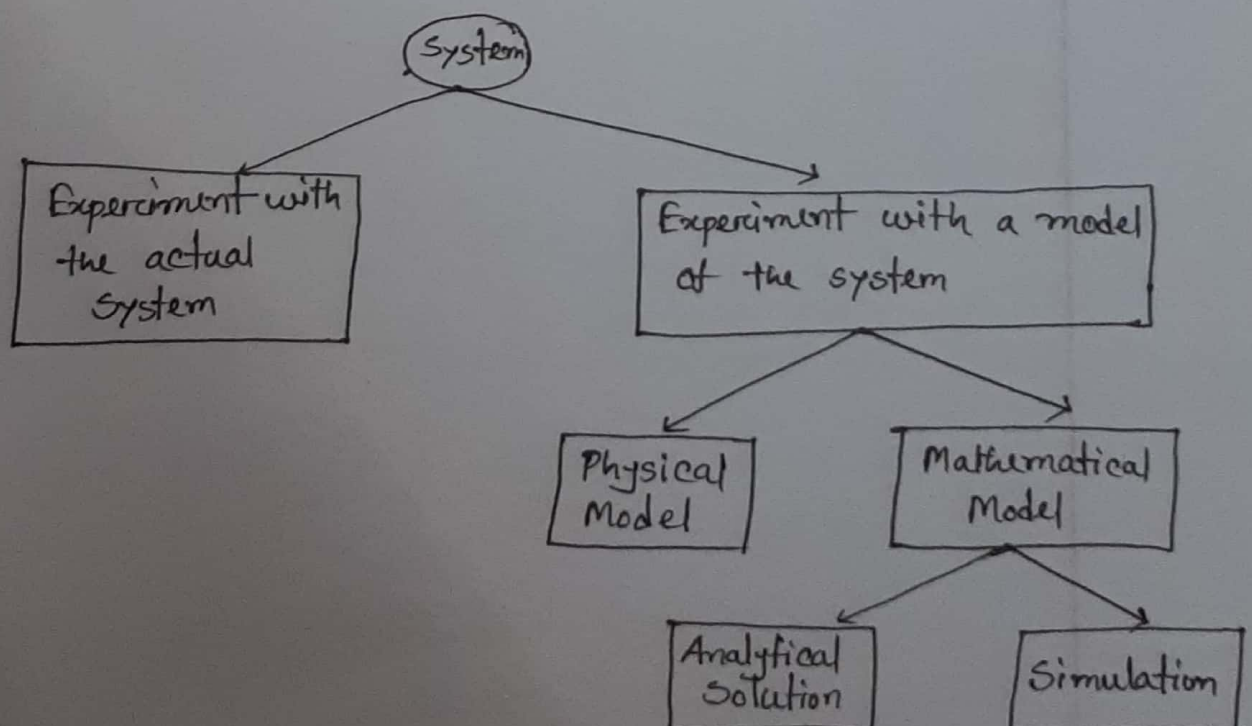
System - is defined to be a collection of entities that act and interact together toward the accomplishment of some logical end.

Two types -

i) Discrete - variable changes instantaneously at a point of time. Example - bank

ii) Continuously - state variable changes continuously with time. Example - Airplane.

** Ways to study a system



* Discrete-Event Simulation

Example 1.1

** Time-Advance mechanism

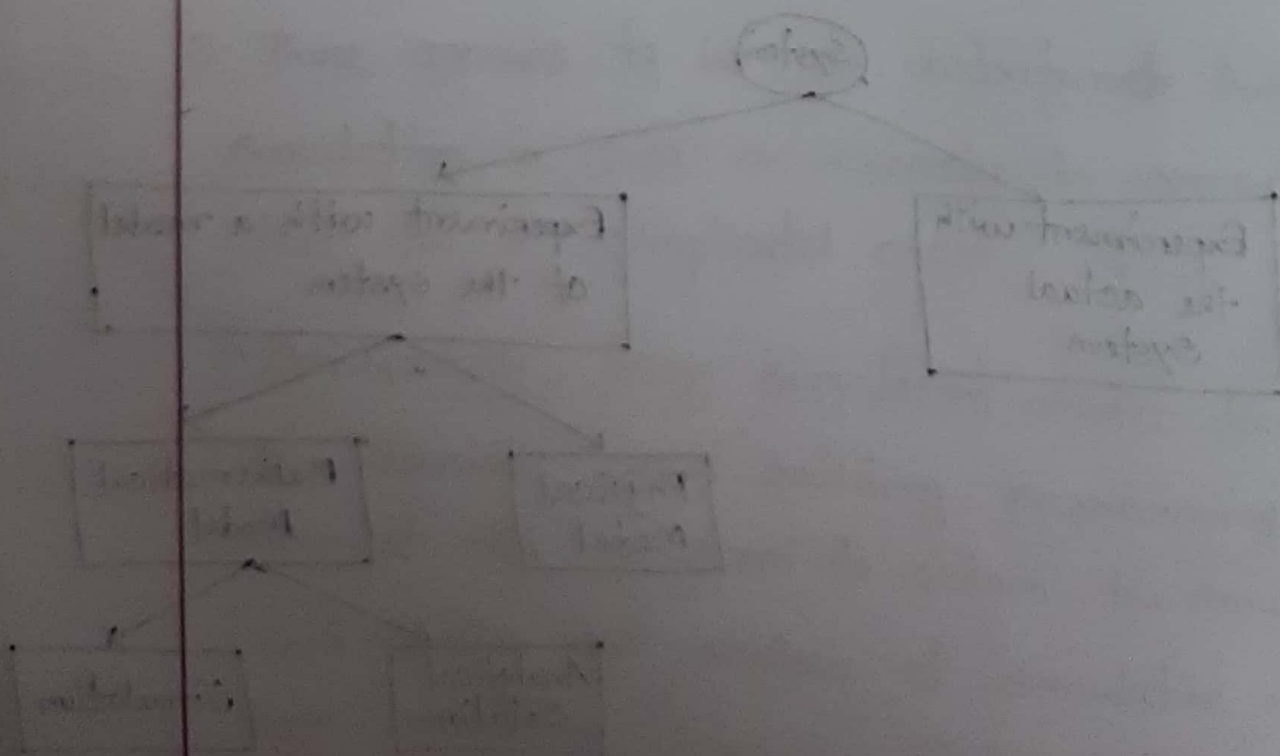
Example 1.2

** Components and Organization of a Discrete-Event Simulation Model.

* Simulation of a Single-server queuing system

Problem statement

Intuitive Explanation



Chapter 04

Review of Basic Probability & Statistics

* Why probability and statistics are needed?

⇒ Probability and statistics are needed to understand how to model probabilistic system, validate the simulation model, choose the input probability distribution, generate random number samples from these distributions, perform statistical analysis of the simulation output data and design the simulation experiments.

** random variables and their properties

Experiment - a process whose outcome is not certain

Random variable - is a function or a rule that assign a real no. to each point in the sample space.

Distribution function -

$$F(x) = P(X \leq x) \quad \text{for } -\infty < x < \infty$$

Properties

Types -

i) Discrete distribution function (countable no.)

ii) Continuous distribution function

Probability mass function - for discrete random variable.

Probability density function - for continuous random variable.

mean $\beta \rightarrow$ parameter

Joint Probability Distribution

Mean (μ) $\rightarrow$ expected value

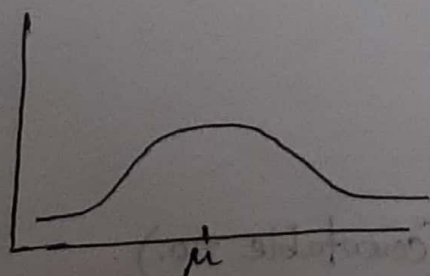
* Why mean/median is needed?

$\Rightarrow$ because exactly statistics to represent

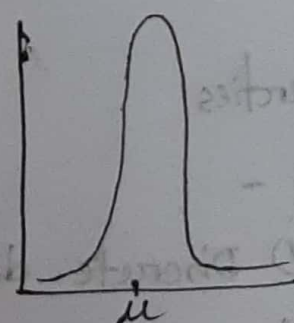
Variance - deviation (data থেকে বড় দূরে আছে)

$$\sigma_i^2 = E[(x_i - \mu_i)^2] = E(x_i^2) - \mu_i^2$$

Gaussian Distribution - bell shaped.



Larger variance



Smaller variance

Variance Properties

* Variance is used for standard deviation

- What type of data $\Rightarrow$ continuous
- Dimension of data $\Rightarrow$ one dimensional

Co-variance

- are symmetrical
- 2 dimensional i.e. 2×2 matrix
- diagonal values σ^2

Correlation

$$\rho_{ij} = \frac{c_{ij}}{\sqrt{\sigma_i^2 \sigma_j^2}}$$

* Simulation Output Data

Stochastic process - a collection of similar random variable ordered over time

Discrete time stochastic process $\rightarrow$ uses discrete random variables.

Continuous " " " $\rightarrow$ uses continuous random variables.

Example - 4.19, 4.20

Co-variance stationary (over time & probabaility output pattern change not; means constant)

- certain properties

- $\mu_i = \mu$
 $\sigma_i^2 = \sigma_i^2$ } mean & variance change not

- multiple dimension (more than 1)

** Intercarrrival time -

Service time -

Intercarrrival time + service time $\rightarrow$ input

Delay time $\rightarrow$ output

$\rho = 5$ बला याव वा, cause $\rho < 1$

$$\rho = \frac{\lambda}{\omega} < 1$$

λ = arrival rate at unit time

ω = service time

$\rho = \frac{\lambda}{\omega} = 1$ means, यात्रा वलाह अवारेकने service एका रहेह.

Correlation

Figure 4.10

* Estimation of means, variance and correlations

$$\bar{X}(n) = \frac{\sum_{i=1}^n x_i}{n}$$

unbiased अयाव वगल expected, $E[\bar{X}(n)] = \mu$

Figure 4.12, 4.13

* Confidence intervals and hypothesis tests for mean

hypothesis - assumption

Central limit theorem - central tendency (mean व वगलकहि अब data वस कृता रहे)

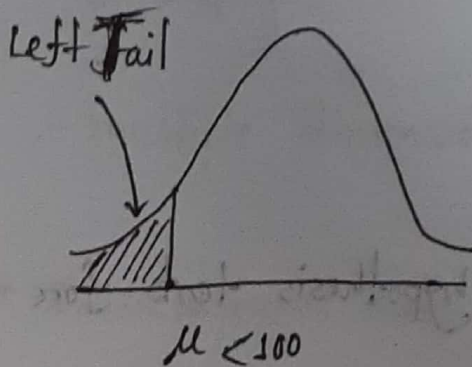
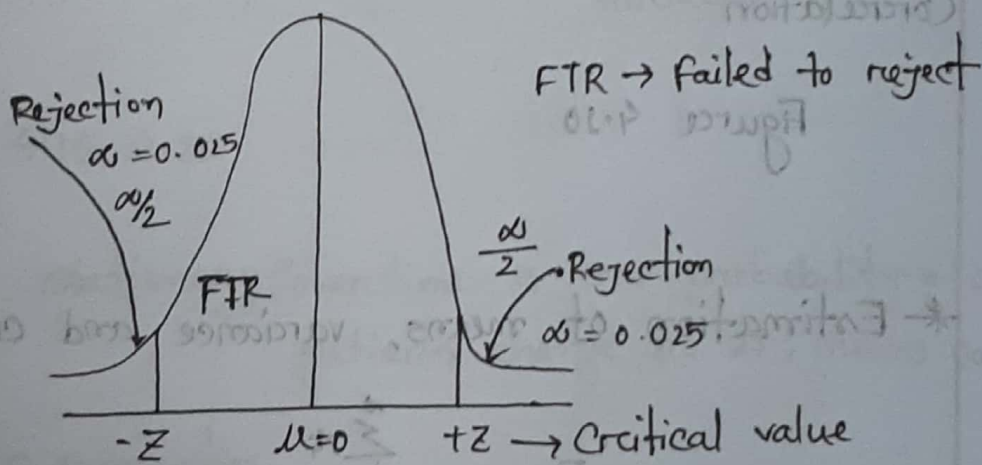
Theorem 4.1

mean = 0 ; means one positive, another side negative

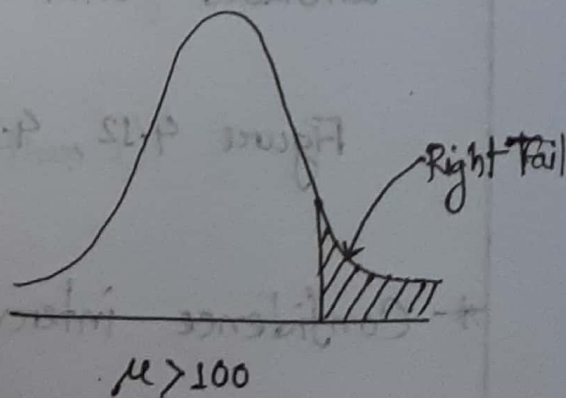
$\mu=0, \sigma^2=1 \rightarrow$ standard normal random variable

$\phi(z) \rightarrow$ standard normal random variable function

Fig - 4.15



(a)



(b)

Hypothesis:

i) Null Hypothesis (H_0): $E_x - \mu = 0$

ii) Alternate Hypothesis (H_a): $E_x - \mu \neq 100$

Problem statement

Solution $\rightarrow$ test statistics, value- α আর compare করে

$C = 0.95$ means 95% confidence level

α = significant level

$\alpha = 0.05$

Z_c compute করে

T-dist $\rightarrow$ special case of normal distribution

Rules $\rightarrow$ যা define করে

i) No. of sample, n এর value < 30 হলে

population standard deviation unknown হলে $\rightarrow$ T-distribution

$$t = \frac{\bar{X} - \mu}{S/\sqrt{n}} \quad \begin{array}{l} n < 30 \\ \sigma = \text{unknown} \end{array}$$

ii) $n > 30 \rightarrow$ normal distribution

$Z_c =$

Example - 4.27

2 Types errors

Table 4.2

i) $\alpha \rightarrow$ if one reject H_0 when in fact its true

ii) $\beta \rightarrow$ if one failed to reject, but it was false

Example 4.29

The Strong Law of large number

$$\bar{X}(n) \rightarrow \mu \text{ w.p. } 1 \text{ as } n \rightarrow \infty$$

Theorem 4.2

Example - 4.31

Figure - 4.38

$$n > \pi$$

$$n - \bar{X}$$

$$\frac{n - \bar{X}}{\sqrt{n}}$$

Chapter 06

Selecting Input Probability Distribution

6.2 Useful Distribution Probability Distribution

Parameterize $\rightarrow$ to define

Parameterization of Continuous Distribution

If the parameters are defined correctly, the probability distribution function density function can be classified, on the basis of their physical or geometric interpretation, as being one of three basic types:

1. Location parameter (γ): specifies an abscissa (x -axis) location point of a distribution's range values. Usually γ is the mid point (μ for normal distribution) or lower endpoint of the distribution's range.

It is also called shift parameters as γ changes, the associated distribution merely shifts left or right without otherwise changing.

$$r.v. X, \gamma = 0$$

$$\text{Then } r.v. Y = X + \gamma$$

Y has a location parameter γ .

2. Scale parameter (β): determines the scale (or unit) of measurement of the values in range of the distribution. The standard deviation σ is a scale parameter for the normal distribution.

A change in β compresses or expands the associated distribution without altering its basic form.

r.v. X , $\beta = 1$

Then r.v. $Y = \beta X$

Y has scale parameter of β .

3. Shape parameter (α): determines the basic form or shape of a distribution within the general family of distributions of interest.

A change in α generally alters a distribution's properties more fundamentally than a location or scale.

Some distributions don't have a shape parameter like exponential and normal.

While others like beta may have two.

From Table 6.3

i) Uniform Distribution

ii) Exponential Distribution

iii) Normal Distribution

Linear Congruential Generator (LCG)

recursive formula

$$Z_i = (aZ_{i-1} + c) \bmod m$$

parameters
of LCG

Z_0 = the seed or starting value
 c = the increment
 a = the multiplier
 m = the modulus

should satisfy

$$m > 0$$

$$c < m$$

$$m > 2$$

Chapter 07

Random Number Generators

Random variates - random variable to value; particular outcome of a random variable.

7.2 Linear Congruential Generator (LCG)

A sequence of integers $Z_1, Z_2, \dots$ is defined by the recursive formula

$$Z_i = (aZ_{i-1} + c) \bmod m$$

where m = the modulus

a = the multiplier

c = the increment

Z_0 = the seed or starting value.

} non-negative integers

should satisfy $0 < m$

$$a < m$$

$$c < m$$

$$Z_0 < m$$

Example - 7.2 : LCG defined by $m=16$, $a=5$, $c=3$, $Z_0=7$.

Soln:

$$Z_i = (5Z_{i-1} + 3) \bmod 16 \quad U_i = \frac{Z_i}{m}$$

<u>i</u>	<u>Z_i</u>	<u>U_i</u>
0	7	—
1	6	0.375
2	1	0.062
3	8	0.500
4	11	0.6875
5	10	0.625
6	5	0.312
7	12	0.750
8	15	0.9375
9	14	0.875
10	9	0.562
11	0	0
12	3	0.187
13	2	0.125
14	13	0.812
15	4	0.250
16	7	0.437
17	6	0.375

Chapter 08

Generating Random Variates

8.2 General Approaches to Generate Random variable Variates:

Inverse Transform

Suppose that we wish to generate a random variate X that is continuous and has distribution function F that is continuous and strictly increasing when $0 < F(x) < 1$.

Let F^{-1} denote the inverse of the function F . Then an algorithm for generating a random variate X having distribution function F is as follows:

1. Generate $U \sim U(0, 1)$
2. Return $X = F^{-1}(U)$

$F^{-1}(U)$ will always be defined, since $0 \leq U \leq 1$ and the range of F is $[0, 1]$.

The illustration of this algorithm graphically is given below where the random variable corresponding to this distribution function can take on either positive or negative values.

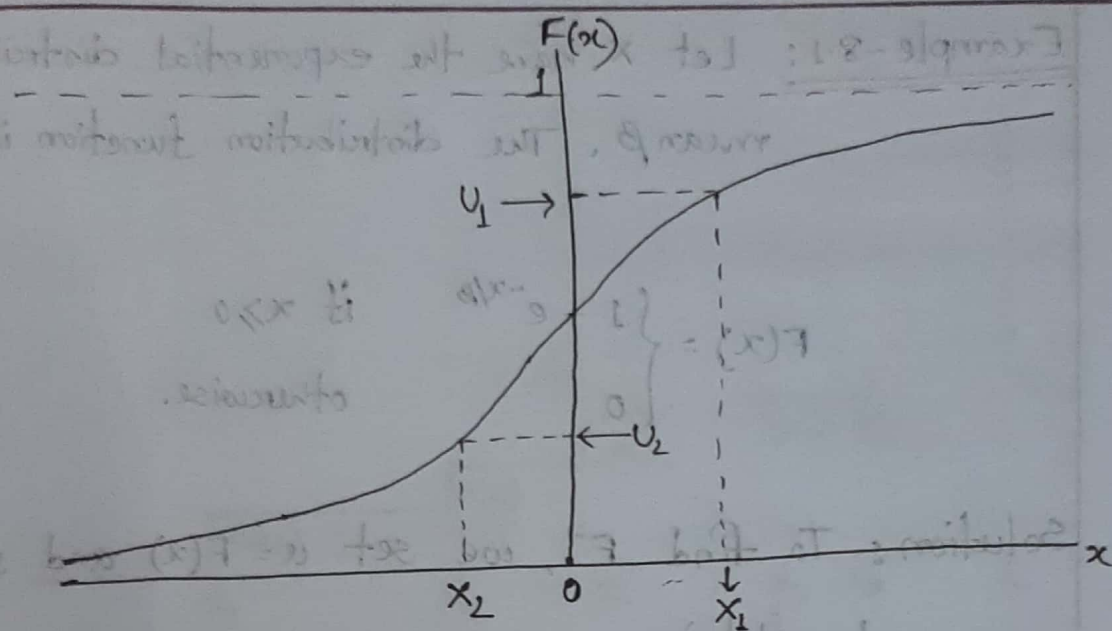


Figure: Inverse transform method for continuous random variables.

In the figure, random number U_1 results in the positive random variate X_1 and the random number U_2 results in the negative random variate X_2 .

For any random real number x , $P(X \leq x) = F(x)$.

Since F is invertible, we have

$$P(X \leq x) = P(F^{-1}(U) \leq x)$$

$$= P(U \leq F(x))$$

$$= F(x)$$

where the last equality follows since $U \sim U(0,1)$ and $0 \leq F(x) \leq 1$.

Example-8.1: Let X have the exponential distribution with mean β . The distribution function is

$$F(x) = \begin{cases} 1 - e^{-x/\beta} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Solution: To find F^{-1} , we set $u = F(x)$ and solve for x to obtain

$$F^{-1}(u) = -\beta \ln(1-u)$$

Thus to generate the desired random variate, we first generate a $U \sim U(0,1)$ and then let $X = -\beta \ln U$.

** When X is discrete, the distribution function is,

$$F(x) = P(X \leq x) = \sum_{x_i \leq x} P(x_i)$$

where $P(x_i)$ is the probability mass function

$$P(x_i) = P(X = x_i)$$

The algorithm is then as follows:

1. Generate $U \sim U(0,1)$
2. Determine the smallest positive integer I such that $U \leq F(x_i)$ and return $X = x_I$

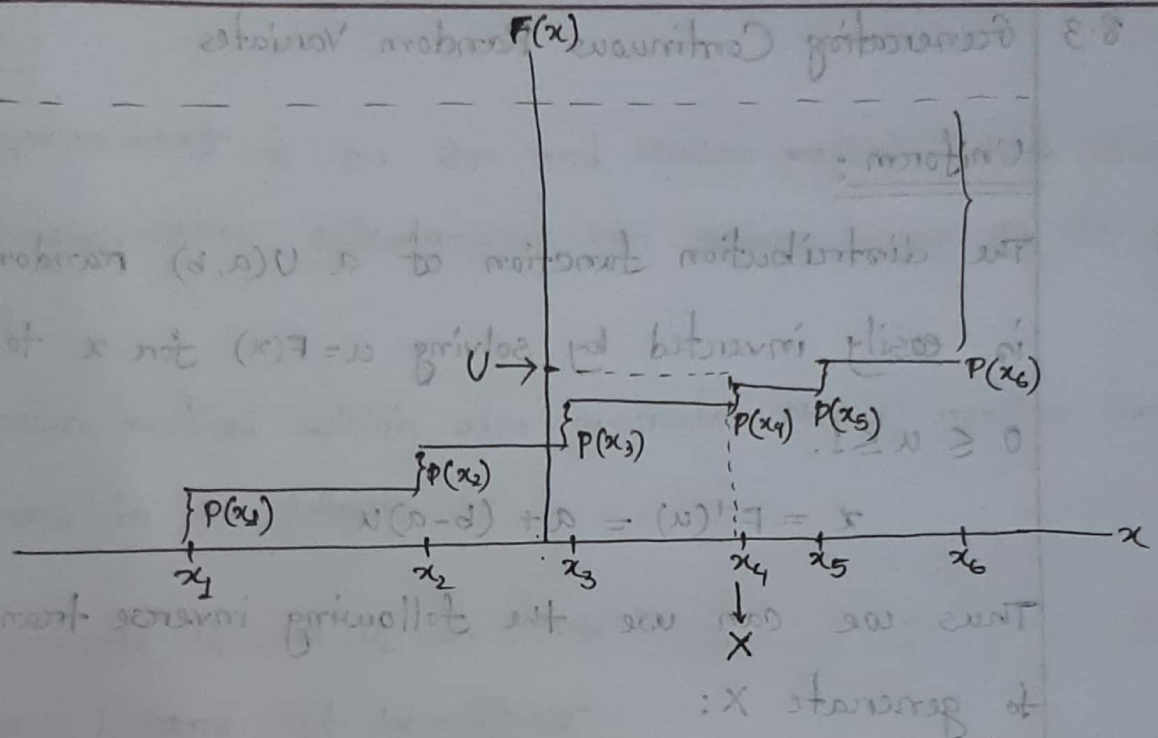


Figure: Inverse-transform method for discrete random variable.

2. Return $X = a + (b-a)U$

Exponential:

1. Generate $U \sim U(0,1)$

2. Return $X = -\ln U$

8.3 Generating Continuous Random Variates

Uniform:

The distribution function of a $U(a, b)$ random variable is easily inverted by solving $u = F(x)$ for x to obtain for $0 \leq u \leq 1$.

$$x = F^{-1}(u) = a + (b-a)u$$

Thus we can use the following inverse transform method to generate X :

1. Generate $U \sim U(0, 1)$
2. Return $X = a + (b-a)u$

Exponential:

The exponential random variable with mean $\beta > 0$, we use the following inverse transform algorithm to generate X :

1. Generate $U \sim U(0, 1)$
2. Return $X = -\beta \ln U$

In order to make the correlation positive, U is used instead of $1-U$.

Normal:

An improvement to the Box and Muller method, which eliminates the trigonometric calculations has become known as the polar method.

The polar method which also generates $N(0,1)$ random variates in pairs is as follows:

1. Generate U_1 and U_2 as IID $U(0,1)$; Let $V_i = 2U_i - 1$ for $i = 1, 2$; and let $W = V_1^2 + V_2^2$
2. If $W > 1$, go back to step 1. Otherwise, let $Y = \sqrt{(2 \ln W)/W}$, $X_1 = V_1 Y$ and $X_2 = V_2 Y$. Then X_1 and X_2 are IID $N(0,1)$ random variates.

Since a rejection of U_1 and U_2 can occur in step 2, the polar method will require a random number of $U(0,1)$ random variates to generate each pair of $N(0,1)$ random variates.